

# Physics Worksheet

Derivative Practice

**Name:** _____ **Student #:** _____ **Student Email #:** _____

1. **Quotient Rule**
   Compute: $$\frac{d}{dx}\left(\frac{\cos x}{\sin x}\right)$$

2. **Chain Rule**
   Compute: $$\frac{d}{dx}\ \ln(\cos(x))$$

3. **Product Rule + Chain Rule**
   Compute: $$\frac{d}{dx}(\cos(x)\sin(x) - \frac{1}{2}\sin(2x))$$

4. **Partial Derrivatives**
   You are standing on a mountain who's shape can be modelled as $z = 7 - \frac{1}{3}x^2 - 2y^4$. You are at the point of $(2, 1, \frac{13}{3})$ and wish to descend the mountain as quikcly as possible. What direction (in unit vector form) should you go in?